

Research Idea: GLEAM_J041509-282854

ALMA Thesis Planner (automated pipeline)

Title: *Is the thermal-frequency emission in GLEAM-turnover galaxies more compact? A common-beam, common-physical-scale compactness comparison in a matched sub-sample of the 2023.1.01342.S targets*

One-line statement. Using the project’s own ALMA data (ADS/JAO.ALMA#2023.1.01342.S), measure the size and surface brightness of the millimetre/centimetre continuum emission in a distance- and angular-size-matched sub-sample of ~ 5 GLEAM-turnover and ~ 5 control galaxies, after convolving every image to a common beam and converting to a common physical scale in parsecs, and answer one empirical question with a real null: **is the continuum emission in turnover galaxies measurably more compact and higher in surface brightness than in the controls?**

Why this is the right question. The proposal’s hypothesis is that low-frequency turnovers arise from free-free absorption in *compact* star-forming regions. That makes a prediction about the ALMA data alone — the emission tracing ionized gas should be concentrated into smaller, brighter structures in turnover galaxies. Testing compactness directly avoids converting an integrated volume emission measure $n_e^2 dV$ into a line-of-sight emission measure, and so never inherits the clump-geometry degeneracy that makes a predicted turnover frequency slide across a decade under equally defensible assumptions. The turnover/control classification is taken as given from the proposal: it is the independent variable.

Step zero, before the thesis plan is fixed (not a scheduled task inside it). Query the ALMA Science Archive for 2023.1.01342.S: which executions were delivered, proprietary status and PI-team access, delivered spectral-window centres, and the shortest and longest baselines per target. ALMA Band 1 and Band 3 have nominal receiver coverages of roughly 35–50 GHz and 84–116 GHz — these are **observed-frame receiver-band definitions, not this project’s tuning**; the frequencies actually used must be read off the delivered SPW headers. If Step Zero shows the data are inaccessible to the student, that is a supervisor conversation about changing projects, not a contingency to be engineered around inside this thesis.

Sample, chosen first. The default sample is the turnover galaxies plus a control set matched in distance and angular size — about five per subsample. Additional targets are added only after the full chain (imaging $\rightarrow$ common beam $\rightarrow$ missing-flux audit $\rightarrow$ size and surface brightness $\rightarrow$ comparison plot) has been driven end to end once. Sixteen targets in two bands is the time sink that would otherwise sit ahead of every science result.

Distances, named and carried. Physical-scale conversion needs a distance per galaxy. The distances used must be taken from the source the PI’s target list itself used, recorded explicitly in the thesis with that provenance and its uncertainty — for nearby galaxies, redshift-derived distances carry peculiar-velocity error at the tens-of-percent level, and that error propagates straight onto the compactness axis. No external distance catalog is introduced here, because the description names

none; if the target list does not record distances, obtaining that provenance from the PI is a Step Zero item.

Deliverable (i) — photometry and imaging. Work from pipeline-calibrated measurement sets and QA2 continuum images; re-image in CASA only where delivered products are inadequate, with no self-calibration on the critical path. Identify and flag channels containing bright lines at each target’s redshift before continuum imaging. Measure integrated flux densities with apertures matched between bands, propagating the per-execution absolute flux-scale uncertainty quoted in each QA2 report.

Deliverable (ii) — resolution and missing-flux audit (the methodological backbone, and the thesis’s likely lasting contribution). Band 1 and Band 3 differ by $\sim 2.5\times$ in frequency and hence in native resolution, and delivered configurations will differ target to target. Before any size is compared across bands or across galaxies, uv-taper and restore every image to a *common* beam, then convert to a common *physical* scale in pc using the sourced distances — otherwise a resolution difference masquerades as the compactness signal being hunted. In parallel, compute the maximum recoverable scale from the *actual* shortest baselines in each delivered execution, compare it to each target’s angular size, and either correct or explicitly bound the resolved-out flux for every galaxy. This matters doubly: resolving out diffuse emission biases the sample toward apparent compactness. The audit must demonstrate that any turnover-vs-control difference is not an artefact of differing uv coverage, angular size, or distance.

Deliverable (iii) — the compactness measurement (the thesis). Per band and per galaxy, at the common beam and in physical units: the size of the emitting region (or an upper limit where unresolved, giving a surface-brightness *lower* limit, which points in the informative direction for the hypothesis), the peak and mean surface brightness, and a concentration measure such as the fraction of flux inside a fixed physical radius — the last being the most robust to beam and missing-flux effects. The headline result is these quantities plotted for turnover galaxies versus controls with missing-flux bounds attached.

Pre-stated power, written before unblinding the plot. With ~ 5 per subsample (~ 8 if the sample grows) and heterogeneous uv coverage, only a large effect is detectable. The student must state in advance, from the achieved common beam, the distance spread, and the missing-flux bounds, the *minimum size ratio and surface-brightness ratio* the data could distinguish. An overlapping result is then a bounded statement — “compact structures of the kind required are absent above this contrast level” — rather than an uninformative null dressed up as a publishable one.

Deliverable (iv) — thermal fraction, demoted to a secondary, explicitly assumption-driven quantity. Between the Band 1 and Band 3 frequencies, free-free dominates at the low end while dust rises at the high end and synchrotron is already weak at Band 1; one spectral index cannot separate three components. The description names no infrared data to pin dust, and GLEAM’s optically-thin end cannot be used to pin the synchrotron slope for precisely the turnover galaxies of interest, since there it is contaminated by the absorption under study. Thermal fractions are therefore reported as *illustrative*, computed under a declared fiducial synchrotron slope, a declared dust contribution, and $T_e = 10^4$ K (free-free emissivity depends only weakly on T_e), with the systematic stated as likely comparable to or larger than any turnover-vs-control difference. Nothing in the headline result depends on this decomposition.

Secondary, descriptive only. Whether surface brightness or concentration correlates with the frequency of the GLEAM turnover within the turnover subsample — presented as a description, not as the test.

Cut point. Drop the per-region resolved analysis of individual compact structures in the best-resolved targets, keeping galaxy-integrated sizes, concentrations, and upper limits. The sample was already the small matched set, so this is the only cut needed to absorb an imaging surprise.

Sources used. The proposal’s own ALMA data and the MWA GLEAM photometry the abstract itself names, plus the distances recorded in the PI’s target list with their stated provenance — nothing else.

Rough schedule (one academic year). ~6 weeks: retrieval, CASA training, reproduce one QA2 image end to end. ~8 weeks: imaging, common-beam convolution, and photometry for the matched sub-sample. ~6 weeks: maximum-recoverable-scale and missing-flux audit, plus the pre-stated detectability thresholds. ~6 weeks: sizes, surface brightnesses, concentrations, subsample comparison; add further targets here if time permits. ~6 weeks: resolved extension and writing.